

Euclid's Elements — Book I

Proposition 33

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Formal Reconstruction — Technical Documentation

The straight lines joining equal and parallel straight lines towards the same parts are also themselves equal and parallel.

1 Introduction

Euclid I.33 states that if two equal and parallel straight segments are joined towards the same parts, then the joining segments are themselves equal and parallel. In the notation used here, the hypotheses are

$$AB \parallel CD, \quad AB \cong CD,$$

and the conclusion is

$$AC \parallel BD, \quad AC \cong BD.$$

The proposition is Euclidean rather than neutral. The genuinely Euclidean step is the passage from parallel lines to congruent alternate angles, corresponding to Euclid I.29 and Hilbert's parallel axiom.

The formal reconstruction is particularly instructive because Euclid's phrase "towards the same parts" contains geometric orientation information which must be made explicit in Lean.

2 The Classical Configuration

For the explicit reconstruction we use the configuration occurring in Beeson's formal proof. The cross-joins AD and BC meet internally at a point M :

$$A - M - D, \quad B - M - C.$$

Thus M lies between A and D , and also between B and C . The two given segments satisfy

$$AB \parallel CD, \quad AB \cong CD.$$

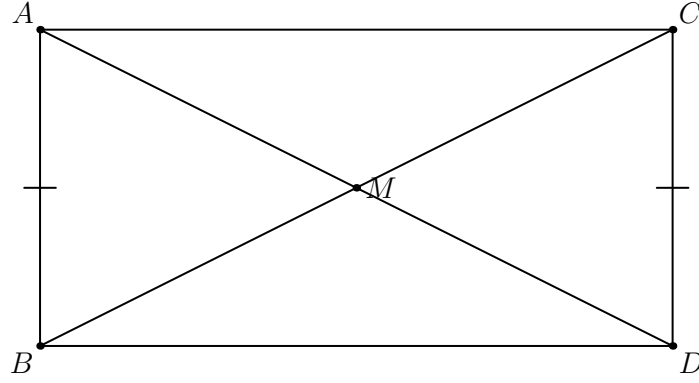


Figure 1: The configuration for Euclid I.33. The segments AB and CD are equal and parallel, while AD and BC meet internally at M .

The required conclusion is

$$AC \parallel BD, \quad AC \cong BD.$$

The betweenness assumptions are not merely a convenient way of drawing the figure. They encode the orientation information hidden in Euclid's phrase "towards the same parts."

3 Classical Proof

The classical dependency chain is short:

$$\text{I.29} \longrightarrow \text{SAS (I.4)} \longrightarrow \text{I.27}.$$

The line BC is a transversal of the parallel lines AB and CD . By I.29 the appropriate alternate angles are congruent.

Together with

$$AB \cong CD$$

and the common side BC , this gives the hypotheses of SAS for the two triangles determined by the configuration.

SAS gives, in particular,

$$AC \cong BD,$$

and also a second pair of congruent alternate angles.

The converse theorem I.27 can then be applied to this second pair of angles, yielding

$$AC \parallel BD.$$

Hence the joining segments are both equal and parallel.

4 The Hidden Orientation Data

The classical diagram makes several facts visually immediate. The formal proof cannot use them implicitly.

From

$$A - M - D, \quad B - M - C,$$

the Lean proof first obtains a line through B, M, C . It then proves that neither A nor D lies on this transversal.

Since M lies on the transversal and is between A and D , the open segment AD meets the transversal. Consequently A and D lie on opposite sides of the line BC .

This produces the formal separation datum

```
hOppositeAD :
HilbertOppositeSide Geo A D trans
```

which is required by the line-level alternate-angle theorem.

The proof also chooses an auxiliary point N between B and C . Thus

$$B - N - C$$

and, by symmetry of betweenness,

$$C - N - B.$$

These relations produce the corresponding same-ray data:

```
hRayBNC :
HilbertSameRay Geo B N C

hRayCNB :
HilbertSameRay Geo C N B
```

They are used to normalize the unoriented angles appearing in the Hilbert interface.

Thus the apparently simple phrase “towards the same parts” expands formally into incidence, betweenness, opposite-side, and same-ray information.

5 Proof Architecture of the Reconstruction

The Lean file contains two levels of I.33.

The first theorem is

```
euclid_proposition_33
```

It is a thin theorem at the level of the developed Hilbert API. Instead of starting from the explicit crossing configuration, it accepts an orientation witness suitable for the existing one-pair-parallel-and-congruent criterion.

The relevant reusable theorem is

```
OnePairParallelCongruentCriterion
```

which recognizes the corresponding quadrilateral as a parallelogram. Equality of the remaining opposite sides is then obtained from

```
ParallelogramOppositeSidesCongruent
```

The second theorem is

```
euclid_proposition_33_crossing
```

This is the configuration-faithful reconstruction. Its hypotheses contain the explicit point M and the two betweenness relations used by Beeson. The proof then reconstructs the classical I.29–SAS–I.27 argument.

The distinction between the two theorems is deliberate. The first shows that the mathematical content of I.33 is already available at a higher level in the Hilbert interface. The second shows how the classical Euclidean configuration is translated into the lower-level data required by that theory.

6 Lean Formalization

The explicit theorem has the following statement:

```
theorem euclid_proposition_33_crossing
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D M : Geo.Point)
  (hParallel : Geo.Parallel A B C D)
  (hCongruent : Geo.Congruent A B C D)
  (hAMD : Geo.Between A M D)
  (hBMC : Geo.Between B M C) :
  Geo.Parallel A C B D /\
  Geo.Congruent A C B D := by
```

The first stage extracts the incidence information associated with $B - M - C$, including a transversal through B, M, C .

Parallelism of AB and CD is then used to prove that A and D cannot lie on this transversal. Together with $A - M - D$, this gives

```
hOppositeAD :
HilbertOppositeSide Geo A D trans
```

An interior point N of BC is introduced to obtain the same-ray normalizations needed by the angle interface.

The Euclidean direction of the alternate-angle theorem is invoked through the following interface theorem:

```
hilbert_alternate_angles_of_parallel_
```

```
oppositeSide_lines
```

After normalization of the rays, this gives the included-angle congruence required for SAS. The two triangles are then compared by

```
SAS
```

and the resulting congruence object is stored as

```
hTriangles :  
TriangleCongruenceResult  
Geo B A C C D B
```

One component gives the required segment congruence:

```
hAC_BD :  
Geo.Congruent A C B D
```

Another component gives the angle congruence from which the second parallelism is obtained. After same-ray normalization the relevant angle statement is

```
hAlternate :  
Geo.AngleCongruent N C A N B D
```

The converse alternate-angle theorem is then applied:

```
hilbert_parallel_of_alternate_angles_  
oppositeSide_lines
```

and yields the required parallelism of AC and BD .

7 Relationship with Earlier Propositions

Proposition I.33 combines two directions of the parallel-angle theory developed immediately before it.

The first direction is the Euclidean one:

parallel lines $\implies$ congruent alternate angles.

This is the content represented by I.29. It is the point at which the Euclidean parallel axiom enters the proof.

The second direction is

congruent alternate angles $\implies$ parallel lines.

This is the converse represented by I.27 and belongs to the neutral part of the theory.

I.33 therefore provides a compact test of the architecture developed for I.27–I.29. The proof begins with the Euclidean direction, passes through SAS, and ends with the neutral converse.

8 Hilbert and Book Zero Components Used

No new axiom is introduced for Proposition I.33.

The explicit proof uses existing infrastructure for:

- incidence and the line through two distinct points;
- betweenness and its incidence consequences;
- separation by a line;
- opposite-side relations;
- same-ray relations;
- segment-congruence symmetries;
- non-collinearity;
- SAS triangle congruence;
- the Euclidean alternate-angle theorem;
- the converse alternate-angle theorem.

The API-level theorem additionally uses the existing parallelogram machinery. In particular:

`OnePairParallelCongruentCriterion`

and

`ParallelogramOppositeSidesCongruent`

No helper specific to Proposition I.33 is added to `HilbertInterface` or to Book Zero.

9 Formalization Notes

The main formal issue in I.33 is not SAS itself. It is the orientation of the configuration.

In an ordinary diagram it is immediately apparent which sides of the transversal contain A and D , and which rays at B and C determine the relevant angles. None of this information is implicit in the formal language.

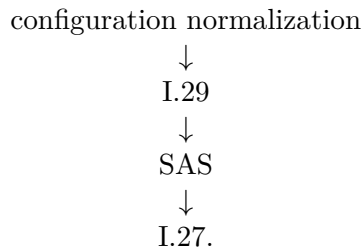
The crossing formulation makes the required information explicit through

$$A - M - D, \quad B - M - C.$$

From these two relations the proof reconstructs the separation and ray data required by the Hilbert interface.

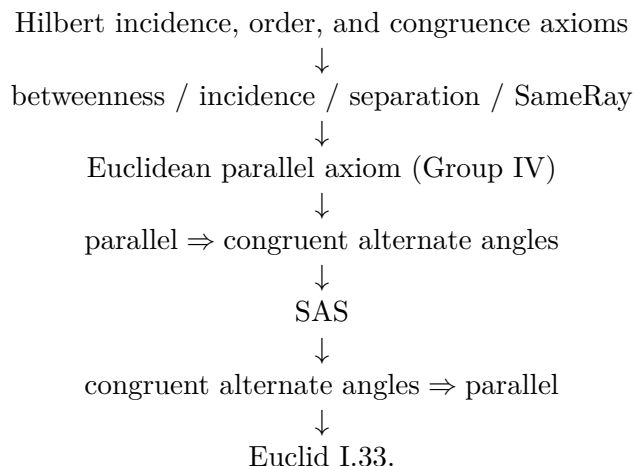
This also explains why the Lean proof is longer than the classical three-step argument. The additional code does not represent new Euclidean mathematics. It reconstructs configuration information which the classical proof obtains from the diagram and from the phrase “towards the same parts.”

The formal proof therefore separates two layers clearly:



10 Dependency Summary

The dependency structure of the explicit reconstruction is:



New axiom: no.

New helper: no.

Build clean: yes.

Sorry: no.

11 Conclusion

Euclid I.33 is mathematically short but formally revealing. Once the orientation hidden in the classical configuration is exposed, the original proof architecture survives:

$$\text{I.29} \longrightarrow \text{SAS} \longrightarrow \text{I.27.}$$

The explicit crossing theorem shows how the classical statement is recovered from betweenness and separation data. The shorter API-level theorem shows, independently, that the underlying parallelogram principle is already available in the developed Hilbert theory.

No new axiom and no Proposition-specific helper are required.